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jakka_bhanu@LAPTOP-DCEV: ~$ cd ~/skill3
jakka_bhanu@LAPTOP-DCEV: ~/skill3$ ls
history_shell  history_shell.c
jakka_bhanu@LAPTOP-DCEV: ~/skill3$ valgrind --version
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
jakka_bhanu@LAPTOP-DCEV: ~/skill3$ sudo apt install valgrind -y
[sudo: authenticate] Password:
Installing:
  valgrind

Installing dependencies:
  gdb          libdw1t64
  libbabeltrace1 libipt2
  libc6-dbg    libsource-highlight-common
  libdebuginfod-common libsource-highlight4t64
  libdebuginfod1t64

Suggested packages:
  gdb-doc gdbserver valgrind-mpi kcachegrind

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 12
  Download size: 27.1 MB
  Space needed: 111 MB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltrace1 amd64 1.5.11-5build1 [165 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight-common all 3.1.9-4.3build2 [64.2 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build2 [273 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 gdb amd64 17.1-2ubuntu1 [4173 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dbg amd64 2.43-2ubuntu2.3 [6407 kB]
Fetched 27.1 MB in 12s (2260 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 42283 files and directories currently installed.)
Preparing to unpack .../0-libdebuginfod-common_0.194-4_all.deb ...
Unpacking libdebuginfod-common (0.194-4) ...
Selecting previously unselected package libdw1t64:amd64.
Preparing to unpack .../1-libdw1t64_0.194-4_amd64.deb ...
Unpacking libdw1t64:amd64 (0.194-4) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../2-libbabeltrace1_1.5.11-5build1_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-5build1) ...
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../3-libdebuginfod1t64_0.194-4_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.194-4) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../4-libipt2_2.1.2-3_amd64.deb ...
Unpacking libipt2 (2.1.2-3) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../5-libsource-highlight-common_3.1.9-4.3build2_all.deb ...
Unpacking libsource-highlight-common (3.1.9-4.3build2) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../6-libsource-highlight4t64_3.1.9-4.3build2_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Selecting previously unselected package gdb.
Preparing to unpack .../7-gdb_17.1-2ubuntu1_amd64.deb ...
Unpacking gdb (17.1-2ubuntu1) ...
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Setting up libipt2 (2.1.2-3) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
jakka_bhanu@LAPTOP-DCEVSCQ0:~/skill3$ valgrind --leak-check=full ./history_shell
==776== Memcheck, a memory error detector
==776== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==776== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==776== Command: ./history_shell
==776==
History-enabled shell. Type commands, use Up/Down arrows to recall.
Type 'exit' to quit.

myshell> test1
You typed: "test1" (history now has 1 command(s))
myshell> test2
You typed: "test2" (history now has 2 command(s))
myshell> exit
Exiting shell...
==776==
==776== HEAP SUMMARY:
==776==     in use at exit: 0 bytes in 0 blocks
==776==   total heap usage: 4 allocs, 4 frees, 1,068 bytes allocated
==776==
==776== All heap blocks were freed -- no leaks are possible
==776==
==776== For lists of detected and suppressed errors, rerun with: -s
==776== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
jakka_bhanu@LAPTOP-DCEVSCQ0:~/skill3$ |
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